

Writing and Solving Linear Inequalities

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

Method

Balance both sides like an equation Undo +/- first, then $\times/\div$ Flip the sign if $\times$ or $\div$ by a negative

Write the answer as an inequality Check with a value from the solution

Common mistakes

- Not reversing the sign when $\div$ or $\times$ by a negative.
- Reversing when only adding or subtracting.
- $<$ and $\leq$ are not the same: $\leq$ includes the end value.
- Writing $x = \dots$ instead of $x < \dots$

Worked example

Solve $3x + 4 < 19$.

Subtract 4: $3x < 15$. Divide by 3: $x < 5$.

Check $x = 4$: $16 < 19$

Visual aid

Visual Example



Number line for $x < 5$: an open circle at 5 (5 not included) with the line shaded to the left to show every value less than 5.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Solve: a) $x + 3 > 7$ b) $x - 2 \leq 5$

Q2 of 8

[2] ● ○ ○ ○ ○

Solve: a) $2x < 10$ b) $5x \geq 20$

Q3 of 8

[3] ● ● ○ ○ ○

Solve: a) $2x + 1 > 9$ b) $3x - 4 \leq 11$ c) $4x + 5 < 5$

Q4 of 8

[3] ● ● ○ ○ ○

Solve (reverse the sign when $\div$ by a negative):
a) $-x > 3$ b) $-2x \leq 8$ c) $5 - x < 2$

Q5 of 8

[3] ● ● ● ○ ○

Solve: a) $3(x - 2) \leq 9$ b) $2(x + 1) > x + 5$ c) $4x - 7 \geq 2x + 3$

Q6 of 8

[3] ● ● ● ○ ○

Form and solve: a) Sam has £ x ; after spending £12 has less than £20. b) Taxi £4 + £2/mile, cost $\leq$ £20 — most whole miles?

Q7 of 8

[3] ● ● ● ● ○

a) Solve $4x + 3 \leq 19$; smallest integer x ? b) Solve $2x - 5 < 7$; largest integer x ?

Q8 of 8

[5] ● ● ● ● ●

Rectangle width x , length $(x+4)$, perimeter < 36 cm, x a positive integer. a) Inequality. b) Solve. c) List possible x .